Reduced-Order Modeling of Upper-Level Winds for Fuel-Efficient Flight Routing

Chennai → Singapore, ERA5 at 250 hPa

AS5401 Final Project Proposal

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Course: AS5401
Date: October 16, 2026

1 Motivation and Project Objectives

(a) Why this problem. At cruise altitude, winds change ground speed by tens of m/s, so the shortest path is not the minimum-fuel path. We study the 250 hPa level (≈ 10.4 km, typical long-haul cruise) over 0°–18°N, 78°–108°E and seek a wind-optimal route Chennai → Singapore (Fig. 1). The problem links a large real dataset (state dimension $n = 17{,}666$) to a practical engineering decision, which makes dimensionality reduction both meaningful and useful.

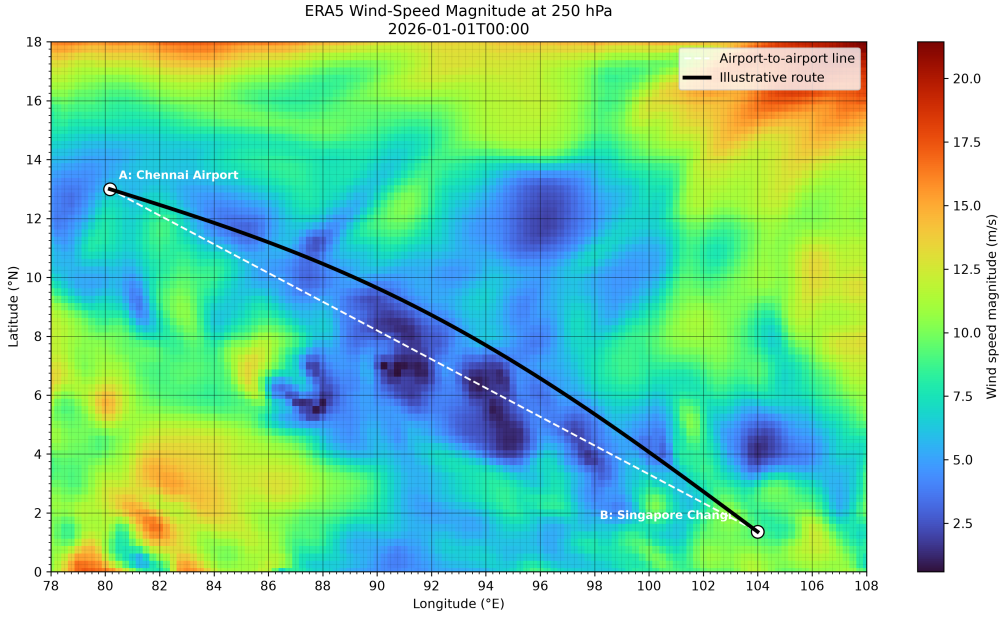


Figure 1: First dataset snapshot with the Chennai–Singapore corridor.

Project structure. The work has two coupled parts:

	Part 1: Wind-field ROM (Main)	**Part 2: Flight-route optimizer(Application)**
Goal	learn low-order models of (u, v) at 250 hPa	use the wind field to find the best route
Input	snapshot matrix $\boldsymbol{X} \in \mathbb{R}^{17666\times 1416}$	wind $\boldsymbol{w}(\lambda, \phi, t)$ from ERA5, POD–DMD or POD–OpInf
Method	POD–DMD, POD–OpInf	time-dependent shortest path (Dijkstra)
Output	$\hat{\boldsymbol{x}}_k = \bar{\boldsymbol{x}} + \boldsymbol{U}_r \hat{\boldsymbol{a}}_k$	route, flight time T, fuel F
Metric	$\varepsilon_{u,v}$, cost	δ_{route}, ε_T, ε_F

The pipeline is

$$\boldsymbol{X} \xrightarrow{\text{POD}} \hat{\boldsymbol{a}}_k \xrightarrow{\text{DMD / OpInf}} \hat{\boldsymbol{x}}_k \xrightarrow{\boldsymbol{C}} \hat{\boldsymbol{w}}(\lambda, \phi, t) \xrightarrow{\text{optimizer}} (\text{route}, T, F), \tag{1}$$

and Part 2 is run on three wind sources (ERA5 reference, DMD, OpInf) so that Part 1 errors can be traced to **route quality**.

(b) Goals of reduced-order modeling.

- *Compression:* replace $n = 17{,}666$ states by $r \ll n$ modal coefficients.
- *Fast wind evaluation:* the route optimizer queries a small model instead of the full data set.
- *Repeated studies:* many departure times and candidate paths at low cost.

(c) Challenges.

- **High dimension and slow singular-value decay:** jet stream, waves and moving features may require many modes.
- **Nonlinearity:** advection is quadratic (Sec. 2).
- **Unclosed dynamics, open domain:** ω, Φ, T are not in the state, and inflow/outflow on all four edges is unmodeled.
- **Short record:** two months of data limit generalization to unseen days.
- **Route sensitivity:** route cost depends only on winds along the path, so local jet errors matter more than global error suggests.

(d) Research question.

> *For equal reduced dimension r, which of POD–DMD and POD–Operator Inference predicts held-out upper-level winds more accurately, and does that accuracy yield a better minimum-time route for Chennai–Singapore?*

2 Governing Equations and Full-Order Model

(a) Variables, inputs, outputs, parameters.

Item	Symbol	Description
States	$u(\lambda, \phi, t),\ v(\lambda, \phi, t)$	zonal / meridional wind at $p = 250\,\text{hPa}$ [m/s]
Coordinates	λ, ϕ, t	longitude, latitude, time (hourly)
Not in state	$\omega = Dp/Dt,\ \Phi,\ T$	vertical velocity [Pa/s], geopotential [m^2/s^2], temperature [K]
Inputs	—	none (autonomous); only the initial state $\boldsymbol{x}_0$
Output	$\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x}$	wind interpolated along the route ($\boldsymbol{C}$: interpolation operator)
Parameters	$f = 2\Omega \sin\phi,\ R,\ c_p,\ g$	$\Omega = 7.292 \times 10^{-5}\,\text{s}^{-1}$, $R = 287\,\text{J/(kg K)}$

(b) Governing equations and physical meaning. The atmosphere is described by the primitive equations [3, 4], here in pressure coordinates and local Cartesian form:

$$\partial_t u + u\,\partial_x u + v\,\partial_y u + \omega\,\partial_p u - fv = -\partial_x \Phi + F_x, \tag{2}$$

$$\partial_t v + u\,\partial_x v + v\,\partial_y v + \omega\,\partial_p v + fu = -\partial_y \Phi + F_y, \tag{3}$$

$$\partial_p \Phi = -RT/p, \tag{4}$$

$$\partial_x u + \partial_y v + \partial_p \omega = 0, \tag{5}$$

$$\partial_t T + u\,\partial_x T + v\,\partial_y T + \omega\,\partial_p T - \tfrac{RT}{c_p p}\,\omega = Q/c_p. \tag{6}$$

Term	Meaning
$\partial_t u,\ \partial_t v$	local change of wind at a fixed point
$u\partial_x + v\partial_y + \omega\partial_p$	advection; the **quadratic nonlinearity**, motivating a quadratic ROM
$-fv,\ +fu$	Coriolis force (Earth's rotation); supports the jet stream
$-\nabla\Phi$	pressure-gradient force from geopotential differences
$F_x, F_y,\ Q$	parametrized friction and heating; small at 250 hPa but non-zero
Eqs. (4)–(5)	hydrostatic balance, mass continuity
Eq. (6)	thermodynamic energy equation

(c) Assumptions and simplifications.

- ERA5 is the reference, not error-free truth (model plus data assimilation).
- Single level: the (u, v) dynamics are not closed; the ROM must learn the missing effects from data.
- Hourly output on a 0.25° grid; faster or smaller scales are unresolved.
- Route study: fixed flight level (no vertical wind). Wind is an *input*, not a decision variable. **Baseline:** constant true airspeed V_a (wind changes only the ground speed V_g), so fuel $F = \dot{m}\,T$ and minimum fuel ≡ minimum time. **Extension:** variable V_a with a speed-dependent burn rate $\dot{m}(V_a)$ (Sec. 4(g)). Neither is a full aircraft performance model.

(d) Initial and boundary conditions.

- **IC:** $\boldsymbol{x}_0$ is the ERA5 snapshot at the start of each evaluation window; the ROMs start from its projection.
- **BC:** the box is a window of a global field (open domain); no boundary condition is imposed by us.

(e) Full-order state dimension and discretization. ERA5 is stored on a regular latitude–longitude grid:

$$N_\phi = \tfrac{18}{0.25} + 1 = 73, \quad N_\lambda = \tfrac{30}{0.25} + 1 = 121, \quad N_g = N_\phi N_\lambda = 8{,}833, \quad n = 2N_g = 17{,}666, \tag{7}$$

$$\boldsymbol{x}(t) = \begin{bmatrix} u_1 & \cdots & u_{N_g} & v_1 & \cdots & v_{N_g} \end{bmatrix}^\top \in \mathbb{R}^n, \qquad \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, t). \tag{8}$$

(f) Sources. Equations follow Holton and Hakim [3] and the IFS documentation [4]; data are from ERA5 [1, 2].

3 Full-Order Simulations and Generated Data

Dataset used. Instead of running our own simulation, we use the public ERA5 reanalysis [1, 2]. *Underlying system:* the global atmosphere. *How generated:* the ECMWF IFS [4] integrates the equations of Sec. 2 and assimilates observations (4D-Var), at a native resolution of about 31 km with 137 model levels; the Climate Data Store interpolates the output to pressure levels on a 0.25° grid. *Not available to us:* the internal model time step, the forcing terms $F_{x,y}, Q$, assimilation increments, and the full 3-D state; only the gridded u, v output is used.

(a) Numerical method and settings.

Quantity	Symbol	Value
Source model	—	ECMWF IFS + data assimilation (ERA5)
Pressure level	p	250 hPa
Domain	ϕ, λ	0°–18°N, 78°–108°E
Spatial resolution	Δs	0.25° (73×121 grid)
Sampling interval	Δt	1 h
Duration	—	1 Jan–28 Feb 2026 (59 days)

For visualization, the speed is $V = \sqrt{u^2 + v^2}$; the ROM state keeps (u, v).

(b) Parameters, initial conditions, inputs. No parameters are varied and no control input is applied. The initial state is the first hourly snapshot, $\boldsymbol{x}_0 = \boldsymbol{x}(t_1)$. For ROM testing, the initial state of each test window is projected onto the POD basis.

(c) Representative plots.

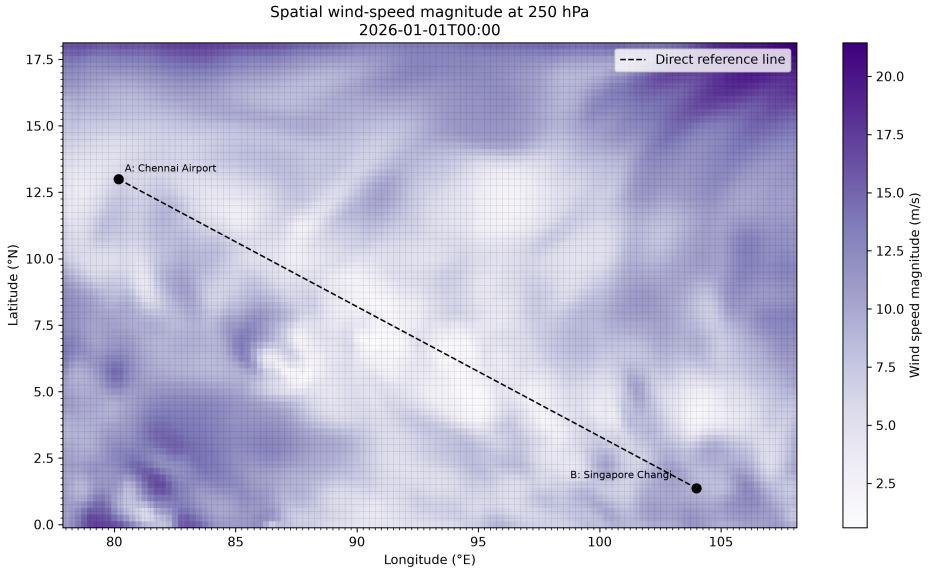


Figure 2: Wind speed $\sqrt{u^2 + v^2}$ [m/s] at 250 hPa, first snapshot (longitude [°E] vs. latitude [°N]).

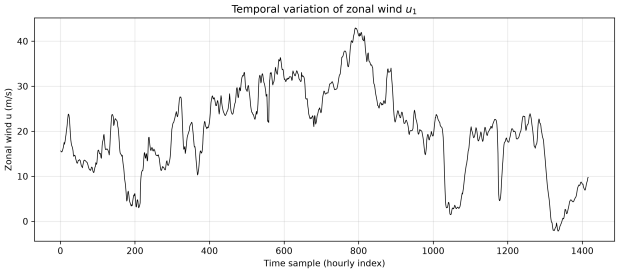


Figure 3: $u_1(t)$ [m/s] at the first grid point.

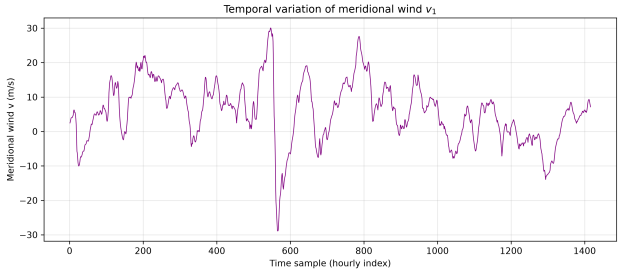


Figure 4: $v_1(t)$ [m/s] at the same point.

(d) Observed behavior and physical reasonableness. The speed field in Fig. 2 is non-negative with spatial structure from the large-scale circulation, while u, v take both signs. The time series in Figs. 3–4 vary on diurnal-to-synoptic scales, as expected for evolving upper-level circulation. Data are screened for missing values, discontinuities and implausible magnitudes before training.

(e) Generated dataset.

Item	Value
Recorded variables	u, v at 250 hPa
Snapshot interval	1 h
Snapshots	$m = 59 \times 24 = 1{,}416$
Simulated cases / windows	1 (single continuous record)
Snapshot matrix	$\boldsymbol{X} \in \mathbb{R}^{17666\times 1416}$, float32, ≈ 95.4 MiB

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{x}(t_1) & \cdots & \boldsymbol{x}(t_m) \end{bmatrix}, \tag{9}$$

where each *row* is one variable (u or v) at one grid point over time, and each *column* is the full wind field at one hour.

(f) Support for the ROM methods; further simulations. The snapshot matrix is the direct input to POD, DMD and Operator Inference. The flow is dominated by large-scale, coherent structures, so a low-dimensional representation is plausible; this will be verified with the singular-value spectrum. No new full-order simulation is needed; the next phase consists of ROM training, held-out prediction and route-optimization runs. Plots, animations and code: https://github.com/Rakshan-VP/era5-wind-rom-flight-routing (`results/`).

4 Proposed ROM Methods and Comparative Study

(a) Methods and justification.

- **POD–DMD:** a simple linear model that fits sampled data without access to the governing equations. Suitable as a baseline for the weakly nonlinear, quasi-periodic part of the flow.
- **POD–Operator Inference (OpInf):** learns a reduced model with a quadratic term, matching the quadratic advection structure of Eqs. (2)–(3).

Both share the same POD basis, so differences come only from the learned dynamics.

(b) Intrusive or non-intrusive.

	POD–DMD	POD–OpInf
Type	non-intrusive	non-intrusive
Access required	ERA5 snapshots only	ERA5 snapshots only
Reduced dynamics	linear	linear + quadratic
Learned parameters	r^2	$r + r^2 + r\,\frac{r(r+1)}{2}$

(c) Construction of the reduced models. Let $\bar{\boldsymbol{x}}$ be the training mean and $\tilde{\boldsymbol{X}}_{\mathrm{tr}} = \boldsymbol{X}_{\mathrm{tr}} - \bar{\boldsymbol{x}}\mathbf{1}^\top$.

$$\tilde{\boldsymbol{X}}_{\mathrm{tr}} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top, \qquad \boldsymbol{U}_r = \boldsymbol{U}_{:,1:r} \in \mathbb{R}^{n\times r}, \qquad \hat{\boldsymbol{a}}_k = \boldsymbol{U}_r^\top(\boldsymbol{x}_k - \bar{\boldsymbol{x}}), \qquad \boldsymbol{x}_k \approx \bar{\boldsymbol{x}} + \boldsymbol{U}_r\,\hat{\boldsymbol{a}}_k. \tag{10}$$

With $\boldsymbol{A}_1 = [\hat{\boldsymbol{a}}_1, \ldots, \hat{\boldsymbol{a}}_{m_{\mathrm{tr}}-1}]$ and $\boldsymbol{A}_2 = [\hat{\boldsymbol{a}}_2, \ldots, \hat{\boldsymbol{a}}_{m_{\mathrm{tr}}}]$:

POD–DMD:

$$\tilde{\boldsymbol{A}} = \arg\min_{\boldsymbol{B}} \|\boldsymbol{A}_2 - \boldsymbol{B}\boldsymbol{A}_1\|_F^2 = \boldsymbol{A}_2\boldsymbol{A}_1^+, \qquad \hat{\boldsymbol{a}}_{k+1} = \tilde{\boldsymbol{A}}\,\hat{\boldsymbol{a}}_k, \tag{11}$$

with modes $\boldsymbol{\Psi} = \boldsymbol{U}_r\boldsymbol{W}$ from $\tilde{\boldsymbol{A}}\boldsymbol{W} = \boldsymbol{W}\boldsymbol{\Lambda}$ and growth/frequency $\ln(\lambda_j)/\Delta t$.

POD–OpInf (discrete-time quadratic model):

$$\hat{\boldsymbol{a}}_{k+1} = \boldsymbol{c} + \boldsymbol{A}\,\hat{\boldsymbol{a}}_k + \boldsymbol{H}\,(\hat{\boldsymbol{a}}_k \otimes \hat{\boldsymbol{a}}_k), \qquad \min_{\boldsymbol{O}} \left\|\boldsymbol{D}\boldsymbol{O}^\top - \boldsymbol{A}_2^\top\right\|_F^2 + \gamma\left\|\boldsymbol{O}^\top\right\|_F^2, \tag{12}$$

where $\boldsymbol{D} = [\mathbf{1},\ \boldsymbol{A}_1^\top,\ (\boldsymbol{A}_1 \otimes \boldsymbol{A}_1)^\top]$ (compact quadratic columns) and $\boldsymbol{O} = [\boldsymbol{c}, \boldsymbol{A}, \boldsymbol{H}]$. The linear-only variant ($\boldsymbol{H} = \mathbf{0}$) is also fitted, and the quadratic term is kept only if validation error improves. The state shrinks from $n = 17{,}666$ to r in both cases.

(d) Selection of r. Candidate values follow the cumulative energy

$$E(r) = \frac{\sum_{i=1}^{r} \sigma_i^2}{\sum_i \sigma_i^2} \in \{0.90,\ 0.95,\ 0.99\}, \tag{13}$$

and the final r (and γ) is chosen on validation error. Both methods use the same r, and the error is also reported as a function of r to show its effect.

(e) Training and evaluation data. Chronological split over the 59 days; the evaluation uses a different, later time interval (and therefore different initial conditions).

Set	Days	Snapshots	Purpose
Train	41	984	mean, POD basis, operators
Validation	9	216	choose r, γ
Test	9	216	final free-run evaluation

No test snapshot enters the basis, r or hyperparameters. Free-run prediction starts from $\hat{\boldsymbol{a}}_0 = \boldsymbol{U}_r^\top(\boldsymbol{x}_0 - \bar{\boldsymbol{x}})$, and $\hat{\boldsymbol{x}}_k = \bar{\boldsymbol{x}} + \boldsymbol{U}_r \hat{\boldsymbol{a}}_k$ is computed without future data.

(f) Performance metrics. Let $\hat{\boldsymbol{x}}_k$ be a ROM prediction and $\boldsymbol{x}_k$ the ERA5 truth, $k \in \mathcal{T}_{\text{test}}$.

Metric	Definition
Wind-field error	$\varepsilon_c(k) = \dfrac{\lVert \hat{c}_k - c_k \rVert_2}{\lVert c_k \rVert_2}$, $c \in \{u, v\}$; mean $\bar{\varepsilon}_c = \dfrac{1}{K}\sum_k \varepsilon_c(k)$; also ε_c vs. lead time $k\Delta t$
Route deviation	$\delta_{\text{route}} = \dfrac{1}{N_s}\sum_{i=1}^{N_s} \lVert \boldsymbol{p}_i^{\text{ROM}} - \boldsymbol{p}_i^{\text{ref}} \rVert$ (great-circle distance, equal-arclength samples)
Flight-time error	$\Delta T = T^{\text{ROM}} - T^{\text{ref}}$, $\varepsilon_T = \dfrac{\lvert \Delta T \rvert}{T^{\text{ref}}}$, with T^{ROM} evaluated on true ERA5 winds along the ROM route
Computational cost	fit time, prediction time, storage ($n \times r$ basis plus operators)
Fuel error	$\varepsilon_F = \dfrac{\lvert F^{\text{ROM}} - F^{\text{ref}} \rvert}{F^{\text{ref}}}$, with F^{ROM} evaluated on true ERA5 winds along the ROM route and speed profile

Relevance: $\varepsilon_{u,v}$ measures the physical accuracy of the wind field, while δ_{route} and ε_T measure the effect on the engineering goal; cost measures the benefit of reduction.

(g) Part 2 and comparison against the full-order reference.

Kinematics. Position (λ, ϕ), true airspeed V_a, heading θ (from north), Earth radius R_e:

$$R_e \cos\phi\, \dot{\lambda} = V_a \sin\theta + u(\lambda, \phi, t), \qquad R_e\, \dot{\phi} = V_a \cos\theta + v(\lambda, \phi, t). \tag{14}$$

On a grid edge of length d, unit direction $\boldsymbol{e}$, unit normal $\boldsymbol{n}$ and local wind $\boldsymbol{w} = (u, v)$:

$$V_g(V_a) = \boldsymbol{w} \cdot \boldsymbol{e} + \sqrt{V_a^2 - (\boldsymbol{w} \cdot \boldsymbol{n})^2}, \qquad \Delta t_{\text{seg}} = \frac{d}{V_g}. \tag{15}$$

Case A (baseline): constant V_a.

$$\min_{\text{route}} \; T = \sum_{\text{seg}} \frac{d}{V_g}, \qquad F = \dot{m}\, T. \tag{16}$$

Case B (extension): variable V_a. Let the burn rate be $\dot{m}(V_a) = c_0 + c_3 V_a^3$ (illustrative: base flow plus power $\propto$ drag$\times V_a$ with drag $\propto V_a^2$; coefficients are normalised, only relative comparisons are reported). Fuel alone would favor very slow flight, so a time cost c_t (cost-index style) is added:

$$\min_{\text{route},\ \mathrm{V_a}\in[\mathrm{V_{min}},\mathrm{V_{max}}]} \; J = \sum_{\text{seg}} \left[\dot{m}(V_a) + c_t\right] \frac{d}{V_g(V_a)}, \qquad F = \sum_{\text{seg}} \dot{m}(V_a) \frac{d}{V_g(V_a)}. \tag{17}$$

For each edge the inner minimisation over V_a is a scalar problem, so Dijkstra is still used with edge cost $\min_{V_a}[\dot{m}(V_a) + c_t]\, d/V_g(V_a)$. Edge weights are evaluated at the arrival time, giving time-dependent Dijkstra.

Comparison. The **same** solver, departure time, constraints, $(V_a, \dot{m}, c_t)$ and test windows are applied to three wind sources: full ERA5 (reference), POD–DMD and POD–OpInf, at the same r.

1. Compare $\varepsilon_{u,v}$ of both ROMs (Part 1).
2. Optimize routes on all three winds, for Case A and Case B (Part 2).
3. Report δ_{route}, ε_T, $\varepsilon_F = |F^{\text{ROM}} - F^{\text{ref}}|/F^{\text{ref}}$ and cost.
4. Test whether lower $\varepsilon_{u,v}$ implies lower δ_{route}, ε_T and ε_F.

References

[1] H. Hersbach *et al.*, "ERA5 hourly data on pressure levels from 1940 to present," Copernicus C3S Climate Data Store, 2023. doi:10.24381/cds.bd0915c6.

[2] H. Hersbach *et al.*, "The ERA5 global reanalysis," *Q. J. R. Meteorol. Soc.*, 146(730):1999–2049, 2020.

[3] J. R. Holton and G. J. Hakim, *An Introduction to Dynamic Meteorology*, 5th ed., Academic Press, 2013.

[4] ECMWF, "IFS Documentation, Part III: Dynamics and Numerical Procedures," Reading, UK.